

## Systems Simulation

### Systems Simulation and Its Processes

- The word simulation is derived from the Latin word *simulare*, which means "to pretend" (Merriam-Webster, n.d.).
- Simulation is a safe way of conducting an experiment with a system model. It is a powerful technique for solving a wide variety of problems. However, the results of the simulation entirely depend on the quality of the system model.
- Simulation can also be viewed as the manipulation of a system model in such a way that it operates based on time/space, enabling one to perceive the relations and interactions that would not otherwise be apparent because of the separation or difference in time/space.
- Simulation greatly helps people in understanding problems where errors or faults are not tolerable.
- Simulation generally refers to the computerization of a developed model, which is run overtime to study the implication of the defined interactions of the parts of the system. The system simulation development is also generally iterative in nature.
- Simulation is an operational research tool that can be used to get the best performance from a given system. It helps to improve existing processes or plan a new facility and fills the gap between exact analysis and physical intuition (Chaturvedi, 2010).

#### Advantages of Simulation:

- It provides a deeper understanding of a system.
- It gives a better insight into how resources are performing.
- It allows stakeholders to try out multiple alternatives.
- It saves time and money.
- It can generate virtual environments.
- It generally improves the quality of the analysis.

The performance of complex systems is difficult to predict, either because of high system complexity or the core theory utilized in the development of the system model is not sufficient. The difficulties in handling such problems using conventional mathematical modeling also arise due to the effect of uncertainties; due to dynamic interactions between system components or complex interdependencies among variables in a system (Chaturvedi, 2010).

### Types of Simulation

- **Physical Simulation** – This deals with the experimentation of a physical prototype of a real system. In this simulation, the real scaled input is applied to the model, and its output is compared with the output of a real system (Chaturvedi, 2010).
- **Numerical Simulation** – This is a common type of simulation in line with the systems approach. Numerical simulation of mathematical models gained popularity because it is easy to develop and modify. It is also considered a *sequential simulation* because calculations proceed in a time-based sequence.
- **Digital (Computer) Simulation** – This simulation uses digital computers to simulate equations that describe a particular system. Digital simulation is distinguished by its clarity of representation, and it is characterized by a high degree of automation (McGraw-Hill Dictionary of Scientific & Technical Terms, n.d.).

### Characteristics of A Simulation (Chaturvedi, 2010)

#### Efficiency

- Higher mission availability
- Increases operational system availability
- Transportation avoidance
- Reduces or eliminates expendable costs
- Less procurement and operational costs

#### Effectiveness

- Improves proficiency or performance
- Forms fewer complex activities
- Provides neutral and obstruction services
- Greater observation, assessment, and/or analysis capability

#### Risk Reduction

- Safety of stakeholders
- Less environmental impact
- Machinery and equipment handling

### Simulation Techniques

#### Monte Carlo (Risk Analysis) Simulation

- It is a mathematical simulation technique to generate random sample data based on some known distribution for numerical experiments. This technique is applied to risk quantitative analysis and decision-making problems. This method can be used in situations where estimations are

necessary and uncertain decisions are inevitable. It can also be used for both stochastic and deterministic problems (Tutorialspoint, n.d.).

- A typical Monte Carlo simulation calculates the model hundreds or thousands of times, each time using different randomly selected values. The results are then used to describe the likelihood, or probability, of reaching various results in the simulation model (RiskAMP, n.d.).

#### Important Characteristics:

- Its output must generate random samples.
- Its input distribution must be known.
- Its result must be known while performing an experiment.

#### The Generalized Flowchart of Monte Carlo Simulation:

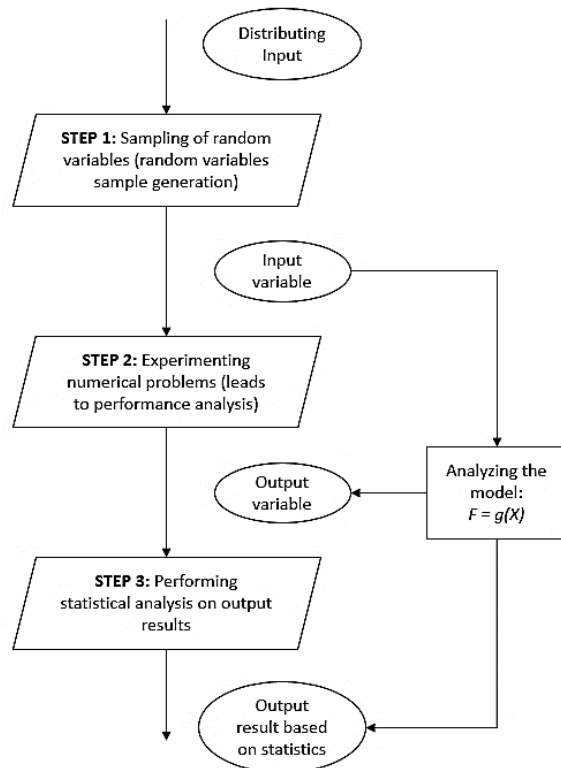


Figure 1. The generalized flowchart of Monte Carlo simulation.

Source: [https://www.tutorialspoint.com/modelling\\_and\\_simulation/modelling\\_and\\_simulation\\_monte\\_carlo\\_simulation.htm](https://www.tutorialspoint.com/modelling_and_simulation/modelling_and_simulation_monte_carlo_simulation.htm)

#### Example:

In a construction project, estimate the time it will take to complete a particular project; based on some expert knowledge, estimate the absolute maximum time it might take, in the worst possible case, and the absolute minimum time, in the best possible case. In this case, it is a construction project with three (3) parts. The parts must be done one after the other. Thus, the total time for the project will be the sum of the three (3) parts. Note that the unit of time is in months (RiskAMP, n.d.).

**Note:** This example uses RiskAMP, a Monte Carlo simulation engine that works with Microsoft Excel. For more information, kindly visit <http://www.riskAMP.com>.

| Task         | Time Estimate    |
|--------------|------------------|
| Job 1        | 5 months         |
| Job 2        | 4 months         |
| Job 3        | 5 months         |
| <b>Total</b> | <b>14 months</b> |

Table 1. A basic forecasting model for a construction project (expected case).

- By using a range of possible values, instead of a single guess, you can create a more realistic picture of what might happen in the future.
- When a model is based on ranges of estimates, the output of the simulation will also be a range.
- When each part has a minimum and maximum estimate, we can use those values to estimate the total minimum and maximum time for the project.
- Based on **Table 1** (expected case), we create a single estimate for each of the three (3) parts of the project, giving us the result of 14 months in total. It might be a good estimate, but the model can't tell us anything about risk. Now this question arises, "How likely is it that the project will be completed on time?"
- To create a model that can be used in a Monte Carlo simulation, we can create three (3) estimates for each job.

| Task         | Minimum          | Most Likely      | Maximum          |
|--------------|------------------|------------------|------------------|
| Job 1        | 4 months         | 5 months         | 7 months         |
| Job 2        | 3 months         | 4 months         | 6 months         |
| Job 3        | 4 months         | 5 months         | 6 months         |
| <b>Total</b> | <b>11 months</b> | <b>14 months</b> | <b>19 months</b> |

Table 2. A forecasting model using range estimates.

- For each job in **Table 2**, we estimate the minimum and maximum expected time based on experience, expertise, or historical information. Then, the 3<sup>rd</sup> column contains the "Most Likely" estimates from **Table 1**.
- Now, **Table 2** contains a range of possible outcomes, indicating that the project may be completed in as fast as 11 months or as late as 19 months. With this, it is possible to describe some of the characteristics of the risk in the model.
- In the Monte Carlo simulation, we will randomly generate values for each job. Then, calculate the total completion time.
- Let us assume that the simulation will be run 500 times.
- To test the likelihood of a particular result, we count how many times the model returned that particular result in the simulation. In this case, we want to know how many times the result was less than or equal to a particular number of months.

| Time      | Number of Occurrences<br>(out of 500 simulations) | Percent of Total |
|-----------|---------------------------------------------------|------------------|
| 12 months | 1                                                 | 0.2%             |
| 13 months | 31                                                | 6.2%             |
| 14 months | 171                                               | 34.2%            |
| 15 months | 394                                               | 78.8%            |
| 16 months | 482                                               | 96.4%            |
| 17 months | 499                                               | 99.8%            |
| 18 months | 500                                               | 100%             |

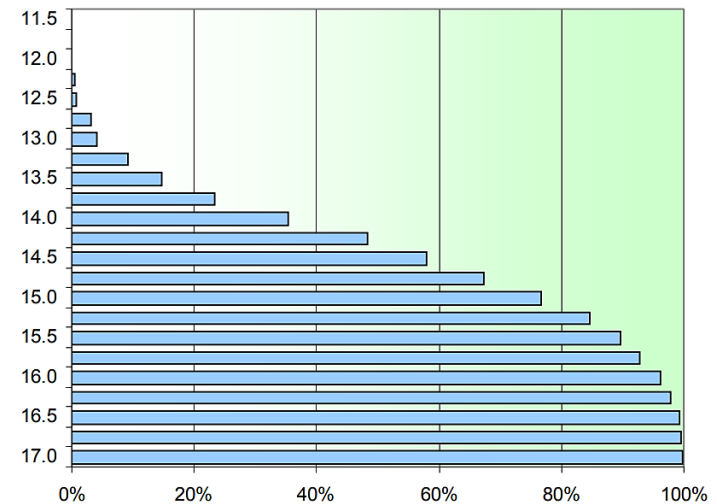
**Table 3.** Possible results of a Monte Carlo simulation.

- The original estimate for the "Most Likely" (expected case) was 14 months. However, in the Monte Carlo simulation, we can see that out of 500 simulations using random values, the total number of occurrences of 14 months or less is only 34.2% of the cases.

#### Simulation Results:

Based on the simulation, there is only a 34.2% chance that any individual trial will result in a total of 14 months or less. On the other hand, there is a 78.8% chance that the project will be completed in 15 months. Furthermore, the simulation model demonstrates that it is extremely unlikely that the project will fall at the absolute minimum or maximum total time estimates.

- The simulation model in **Table 3**, in fact, demonstrates risks. Different choices might impact the project development. Having more information about the risks (other variables associated with the project) at the beginning of the simulation process can contribute to more accurate results and a better plan moving forward.



**Figure 2.** A representation of the probability of project completion within a specific time frame (in months) using RiskAMP.

Source: <https://www.riskamp.com/files/RiskAMP%20-%20Monte%20Carlo%20Simulation.pdf>

#### Agent-Based Simulation

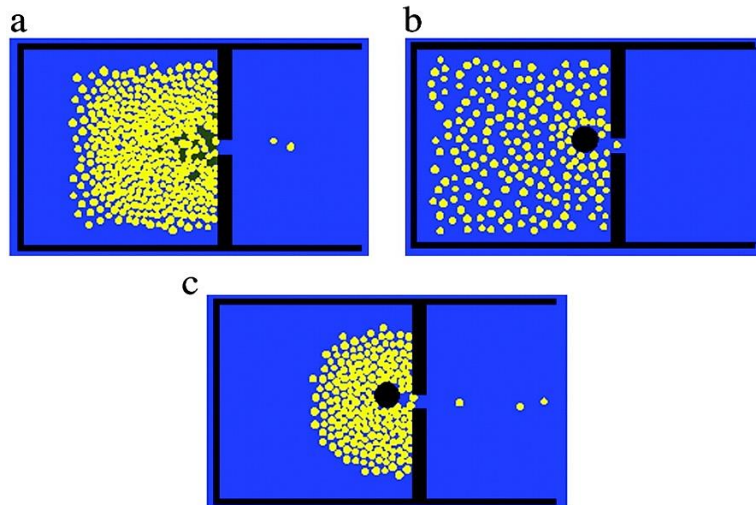
- This focuses on the individual active components of a system. This technique contrasts with both the abstract system dynamics approach and the process-focused discrete event method. The active entities relevant to the system (such as people, vehicles, equipment, or products) must be identified, and their behavior must be defined. The connections between each entity must be established, and the environmental variables must be set (AnyLogic, n.d.). This type of simulation can be used when (Bonabeau, 2002):
  - The interactions between entities are complex, non-linear, or discontinuous;
  - The space is vital, and the entities' positions are not fixed;
  - The population is heterogeneous, and each individual is potentially different;

- The topology of the interactions is heterogeneous and complex; and
- The entities exhibit complex behavior and learning adaptation.

Example:

Crowd stampedes induced by panic often lead to fatalities as people are crushed or trampled. Such phenomena may be triggered in life-threatening situations such as fires in crowded buildings, or may arise from the rush for seats, or sometimes seemingly without any cause. For this example, let us consider a fire escape situation, with 200 people, in a confined space, such as a movie theater or a concert hall, with only one exit point available (Bonabeau, 2002). How can we increase the outflow of people?

- To narrow down the problem, we can ask: *What is the effect of placing a pillar just before the exit, slightly asymmetrically (to the left of the exit point), and about one (1) meter away?*
- Intuitively, one might think that the pillar will slow down the outflow of people. However, agent-based simulation, backed by real-world experiments, indicates that the pillar regulates the outflow of people leading to fewer casualties.



**Figure 3.** A fire escape agent-based simulation (simulation at [www.helbing.org](http://www.helbing.org)). People presented with yellow (not injured) and green (injured) circles in an enclosed area.  
Source: [https://www.pnas.org/content/99/suppl\\_3/7280](https://www.pnas.org/content/99/suppl_3/7280)

### Simulation Results:

(a) No pillar; (b) With pillar, after 10 seconds; (c) With pillar, after 20 seconds. Without the pillar, 44 people escaped, and five (5) people were injured after 45 seconds. With the pillar, 72 people escaped, and no one was injured after 45 seconds.

### Discrete Event Simulation

- This focuses on the processes in a system at a medium level of abstraction. It is widely used in the manufacturing, logistics, and healthcare industries. It is a simulation technique that involves the process of codifying the behavior of a complex system as an ordered sequence of well-defined events. An event comprises a specific change in the system's state at a specific point in time (TechTarget, 2012).

### Important Characteristics:

- There are predetermined starting and ending points, which can be discrete events or instants in time.
- It includes a list of discrete events that have occurred since the process began.
- It includes a list of discrete events which are pending or incoming until the process ends.
- It must track the time that has elapsed since the process began.
- There is a graphical, statistical, or tabular record of the function for which the discrete event simulation is currently engaged.

Example:

Consider the automated teller machine (ATM) counter for discrete system simulation. The following components may be considered for this system (Chaturvedi, 2010).

System states:  $Q(t)$  – number of customers in line

$S(t)$  – server status at time  $t$

Entities: customer and server

Events:  $A$  – arrival

$D$  – departure

$E$  – stopping event

Event Notice:  $(A, t)$  – an arrival event to occur at time  $t$

$(D, t)$  – a customer departure at time  $t$

$(E, T_{sim})$  – the simulation stop event at time  $T_{sim}$

Activities: inter-arrival time, service time

Delay: time spent of each customer in the system

▪ Pseudo-code for the arrival and departure event:

```
% Pseudo code for arrival event
Arrival event occur at Clock =t
If S(t) ==1 then
    Q=Q+1;
Else
    S(t) = 1
    Generate service time Ts
    Schedule new departure at t+Ts
End
Generate Inter-arrival time Ta
Schedule next arrival at t+Ta
Collect statistics
    Time advance control routine
Return
% Pseudo code for departure event
departure event occur at Clock =t
If Q(t) >0 then
    Q=Q-1;
```

▪ Possible input data and the simulation results:

|                    |   |   |   |   |   |
|--------------------|---|---|---|---|---|
| Inter-arrival time | 0 | 5 | 3 | 1 | 1 |
| Service time       | 3 | 2 | 1 | 2 | 1 |

**Table 4.** Inter-arrival time and their service time for customers.

Source: Modeling and simulation of systems using MATLAB and Simulink, 2010. p. 614

| States |      |      | Cumulative Statistics |    | Description                                  |
|--------|------|------|-----------------------|----|----------------------------------------------|
| Clock  | Q(t) | S(t) | B                     | MQ |                                              |
| 0      | 0    | 1    | 0                     | 0  | First arrival at Ta = 0<br>Service time = 3  |
| 3      | 0    | 0    | 3                     | 0  | First departure                              |
| 5      | 0    | 1    | 3                     | 0  | Second arrival Ta = 5<br>Service time Ts = 2 |
| 7      | 0    | 0    | 5                     | 0  | Second departure                             |
| 8      | 0    | 1    | 5                     | 0  | Third arrival Ta = 8<br>Service time Ts = 1  |
| 9      | 0    | 1    | 6                     | 0  | Third departure                              |
| 9      | 0    | 1    | 6                     | 0  | Fourth arrival Ta = 9<br>Service time Ts = 2 |
| 10     | 1    | 1    | 8                     | 1  | Fifth arrival Ta = 10<br>Service time Ts = 1 |
| 11     | 0    | 1    | 8                     | 1  | Fourth departure                             |
| 12     | 0    | 0    | 9                     | 1  | Fifth departure                              |

**Table 5.** Simulation results for an ATM counter.

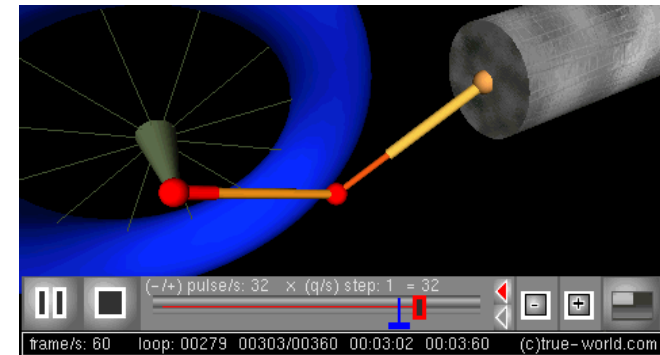
Source: Modeling and simulation of systems using MATLAB and Simulink, 2010. p. 614

## System Dynamics Simulation

- This is a highly abstracted method of modeling and simulation. It ignores the fine details of a system, such as the individual properties of people, products, or events, and produces a general representation of a complex system. The abstract simulation models generated in this technique may be used for long-term strategic modeling and simulation (AnyLogic, n.d.).

Example:

**Figure 4** is a snippet from a 3D modeler in a software system dynamics simulator. The calculated values are associated with the parameters of the rod and piston. In this example, the piston is providing the force for the wheel to rotate, while the speed of rotation, the radius of the wheel, and the length of the rod vary.



**Figure 4.** Piston motion equations in a 3D modeler in a software system dynamics simulator.

Source: [https://en.wikipedia.org/wiki/Dynamic\\_simulation](https://en.wikipedia.org/wiki/Dynamic_simulation)

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